

**MORPHOMETRIC AND POPULATION ANALYSIS OF COLLECTOR URCHINS
(*TRIPNEUSTES GRATILLA*) AND BLACK SEA URCHIN (*DIADEMA ANTILLARUM*)
ON SELECTED COASTAL AREAS OF GINGOOG CITY**

A RESEARCH PAPER

**PRESENTED TO THE FACULTY OF THE DEPARTMENT OF AGRIBUSINESS,
GINGOOG CITY UNITED COLLEGES, IN PARTIAL FULFILMENT
OF THE REQUIREMENTS FOR THE DEGREE**

BACHELOR OF SCIENCE IN AGRIBUSINESS

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December 2024



GINGOOG CITY UNITED COLLEGES

Barangay 22. Motoomull St., Gingoog City, Misamis Oriental

Bachelor of Science in Agribusiness



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ABSTRACT

The study, titled "Morphometric and analysis of Collector Urchins (*Tripneustes gratilla*) and Black Sea Urchin (*Diadema antillarum*) on Selected areas of Gingoog City," aimed to determine the impact of a factory in Barangay San Luis on sea urchin populations. Researchers employed a quadrat sampling technique to assess the morphometric characteristics (spine length and mass) and population of two sea urchin species in two locations: Barangay Agay-ayan (without a factory) and Barangay San Luis (with a factory). The study's significance lies in providing insights into the health and abundance of sea urchin populations in the area, informing policymakers about the impact of industrial pollution, and empowering local communities to engage in environmental stewardship. The findings suggest that while the factory infrastructure in San Luis did not significantly affect overall population, spine length, or mass, it may have influenced species-specific responses, particularly in terms of population distribution. The study underscores the importance of considering species traits and local environmental factors when evaluating anthropogenic impacts on marine ecosystems.

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to the individuals who have played a significant role in the successful completion of this study. First and foremost, we are immensely thankful to our Almighty God for the divine guidance, strength, and blessings that have been our constant source of inspiration throughout this journey. We acknowledge the huge impact of the mercy of our All-Powerful God, who has made this accomplishment possible by guiding our path and providing us the strength to face difficulties.

We would also like to express our appreciation to our parents for their endless support in finance and guidance throughout the process of the completion of the study. Their support made it possible, and it has inspired and motivated us to complete the study. We are immensely thankful for their steadfast presence and support.

We are deeply grateful to our research adviser, Mr. Christian Guillemore O. Susi, for his invaluable guidance and support throughout our research. His encouragement and expertise were instrumental in completing our study. His visit during our sample collection was particularly helpful, as he provided valuable insights into the local species. We are truly fortunate to have had the opportunity to learn from such an exceptional educator.

DEDICATION

This Feasibility Study is dedicated with heartfelt gratitude to our:

Christian Guillemore O. Susi, Crispher Zeus C. Baal, LPT, Jhon Kenneth A. Uban, L. Agr., MSc. (Occ.), Ayesha Louise Llaban, L. Agr. MSc. (Occ.), and Julie G. Bacasnot, LPT MAEd-Math. our esteemed mentor and panelists, for invaluable guidance, unwavering support, and the continuous encouragement that fueled our research journey and during to our defense day. Your expertise and insightful feedback were instrumental in shaping this study.

Our beloved Parents, whose unwavering love, constant support, and unwavering belief in us have been the driving force behind our success. Your sacrifices and encouragement have paved the way for our achievements.

Our cherished classmates and friends, for their valuable insights, helpful discussions, and the camaraderie that made this journey more enjoyable. Your support and understanding provided a strong foundation for our perseverance.

Gingog City United Colleges, our esteemed institution, for providing the necessary resources and a conducive learning environment that fostered our academic growth and facilitated the successful completion of this study.

This study would not have been possible without the contributions and support of each and every one of you.

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INTRODUCTION

Background of the Study

Sea urchins are small, spiky, and round animals (Hall, 2023). They are part of the Echinidae family and live in oceans worldwide. Sea urchins don't swim; they stay on the ocean floor. These creatures play a crucial role in marine ecosystems. As per the Reef Resilience Network (2023), sea urchins eat algae and create space for corals to settle, which is vital in reefs where other plant-eating animals like parrot-fish and rabbit-fish are scarce. The Food and Agriculture Organization of the United Nations (FAO) mentioned that global sea urchin catches reached a peak of about 120,000 tonnes in 1995 but dropped to around 75,000 tonnes in 2022. This decline is due to various reasons, including too much fishing, damage to their habitats, and outbreaks of diseases (FAO, 2023).

The Philippines has a diverse and abundant sea urchin population, with around 230 species living in its waters. This number is higher than the total sea urchin species in the Gulf of Mexico, showing the Philippines' special marine biodiversity (SICB, 2017). A study in 2018 by Cruz and Ebanes found that the most common sea urchin species in the Philippines are the black sea urchin (*Diadema Antillarum*), the green sea urchin (*Strongylocentrotus Nudus*), and the red sea urchin (*Echinometra mathaei*). These sea urchins live in various marine environments, from coral reefs to sea grass beds to sandy areas, and they play important roles in these ecosystems.

The decrease in sea urchin populations varies in different regions. For example, in the Mediterranean Sea, sea urchin numbers have dropped significantly due to overfishing, while in places like the Pacific Northwest, populations have stayed the same or even grown (FAO, 2023). In the Philippines, many people gather sea urchins for their meat, which is rich in protein and omega-3 fatty acids. The most commonly harvested species is the *Tripneustes Gratilla*, the collector urchin. Other edible sea urchins include the red sea urchin (*Strongylocentrotus Franciscanus*), purple sea urchin (*Strongylocentrotus Purpuratus*), and black sea urchin (*Salix fragilis*).

The *Tripneustes Gratilla*, also known as the collector sea urchin, lives in various tropical habitats like coral reefs, sea grass beds, algae meadows, and sandy areas (Lyimo et al., 2011). In these places, they eat different types of algae and sea-grasses (Lyimo et al., 2007). Their bodies are usually dark purple, and they have short spines with sharp ends. Their color can range from white to copper brown or orange (Capinpin Jr., 2015). These sea urchins can grow up to 10-15 cm in diameter and are hunted by puffer fish, octopuses, and humans.

The *Diadema antillarum*, also called Black Sea Urchins or long-spined sea urchins, are known for their long black spines. These sea urchins live in warm waters worldwide, mainly in shallow, rocky areas where there is plenty of food for them to eat. Research on similar species like (*D. Antillarum*) suggests that strong water movement limits their presence in the shallowest rocky shore areas. The spines of (*Diadema antillarum*) species are very thin, making it hard for them to stay attached to surfaces, which makes them less able to handle strong water currents (Tuya, F. et al., 2007).

In Gingoog City, Misamis Oriental, Philippines, farming sea urchins has become a significant way for local communities to get food and make money. Fishers gather sea urchins from different coastal areas, like Brgy. Agay-ayan, which is famous for its diverse marine life and clean surroundings. However, there are worries about a factory in Brgy. San Luis near the sea and how its activities might affect the marine environment nearby.

This research assessed the current state of sea urchin populations in Brgy. Agay-ayan and Brgy. San Luis in Gingoog City during the normal day. The study utilized morphometric analysis to examine the physical characteristics of sea urchin structures, providing valuable insights into the health and abundance of these populations. A systematic and careful evening collection of sea urchin was made on the weeks of October in low tide. Two buckets differentiated the two species, which included Collector Urchins (*Tripneustes gratilla*) and Black Sea Urchin (*Diadema antillarum*). These collections came with the support of flashlights and stainless tongs to facilitate safe handling. The sea urchins were collected carefully by hand, avoiding injury to them, and immediately transferred for inspection. Their mass was weighed using a calibrated digital scale, and the length of the longest spine was measured with a transparent ruler.

Statement of the Problem

The study evaluated the population and morphometric characteristics of Collector Urchins and Black Sea Urchins in selected coastal areas of Gingoog City.

Specifically, the study sought to answer the following:

1. What is the morphometric characteristics of Collector Urchins and Black Sea Urchins in Selected coastal areas of Gingoog City.
2. What is the population of Collector Urchins and Black Sea Urchins in Coastal areas located in Gingoog City.
3. What are the differences in morphometric and population of specified sea urchins in Gingoog City.

Objectives of the Study

This study aims to evaluate the population and the morphometric characteristics of Collector Urchins and Black Sea Urchins in Barangay Agay-ayan and San Luis, Gingoog City.

Specifically, this study aims to:

1. Determine the morphometric characteristic of Collector Urchins and Black Sea Urchins in Brgy. Agay-ayan and Brgy. San Luis in terms of their spine length and mass.
2. Assess the population analysis of Collector Urchins and Black Sea Urchins in selected brgy. in Gingoog City.
3. Evaluate the differences in morphometric characteristics and population of Collector Urchins and Black Sea Urchins in brgy. San Luis and Agay-ayan.

Significance of the Study

This study gave significant insights to many stakeholders. Policymakers got valuable insights about the impact of industrial pollution on sea urchins, hence, in formulating effective environmental regulations and conservation strategies. Local communities of Brgy. Agay-ayan and Brgy. San Luis increased their knowledge of healthy marine ecosystems and consequences of human activities, potential for environmental stewardship. Conservation groups were better equipped to direct their conservation efforts by having a clearer picture of the health of sea urchin populations. Fishermen learned how their practices were affecting sea urchin populations, which enabled them to create sustainable fishing methods. Additionally, current and future researchers were better equipped to understand sea urchin biology and behavior, creating a basis for further research and the development of standardized measurement methods applicable across different marine environments.

Scope and Limitation

The study included the collection of sea urchin samples at Barangay Agay-ayan and San Luis, Gingoog City, Misamis Oriental, Philippines. These two barangays represented the entire coast of Gingoog City, as Gingoog City has only two classified species of sea urchin: Collector Urchins (*Tripneustes gratilla*) and Black Sea Urchins (*Diadema antillarum*). These two species were used in the study. Data collection occurred during natural days and assessed spine length and mass. Start our data sample collection from October 7 – October 13, 2024, 5 days rest interval time starts at 5:00pm until 10:00pm.

REVIEW OF RELATED LITERATURE

A. Factors Affecting Sea Urchin Spine Health and Morphology

The condition of sea urchin spines is closely related to the general state of marine environment conditions and is of significant importance in indicating the health status of marine ecosystems, which is caused by a variety of factors related to spine damage. In addition, according to Nakamura et al. (2012), the chance of sea urchin fertilization success highly depends on spine morphology. It plays a role in protecting sea urchin embryos from predators which could also lead to an increase in the number of sea urchin offspring that survive to adulthood. Sea urchin spines are exposed to several influences that can greatly affect their health and function. Starvation, the result of the lack of algaenous ingestion, causes spine loss, reduced gonad size, and fertility output is severely compromised (Welsh, C. H. n.d.; Sean, B. 2021).

Moreover, acidification or the decrease in seawater pH from excess carbon dioxide absorbed damages calcification, growth, and survival of sea urchins, including their spines. It also interferes with fertilization by affecting sperm movement, egg activation, and embryonic development (Welsh, C. H. n.d.; Towle, E. K. 2022). Warming, in addition, brought about by an intensified greenhouse effect, affects the reproductive cycle, the timing of spawning, and the quality of gametes of sea urchins. Further to the acidification, warming reduces fertilization success and larval survival combined (Welsh, C. H. n.d.; Towle, E. K. 2022). Rahman, B. et al. (2023) provided an extensive study on diversity in sea urchin spines, where the structures are shaped according to the environment and lifestyle. The diversified shapes of sea urchin spines are not random;

they are related to the environmental stresses under which sea urchins are positioned. Sea urchins adopt long, slim spines in seaweed forests for effective blending, and in exposed areas, they develop short, robust 8 spines to better defend against predators. Factors like water movement, substrate type, predator presence, and climate contribute to the diverse characteristics of spines. However, increased atmospheric carbon dioxide levels lead to ocean acidification, negatively impacting the formation and regeneration of sea urchin spines, which are composed of high-magnesium calcite. Ocean acidification, which has been brought about by increased atmospheric carbon dioxide levels as a result of human activities, threatens sea urchins (Emerson, C. E. et al., 2017). Another factor that can affect the sea urchin's spines is the availability of food. Loss of spines is also caused by starvation. Their growth, reproduction, and defense can decrease. According to Tsafnat et al. (2012) starvation can also affect the mechanical properties of the sea urchin spines, and they would turn out to be brittle with a tendency to fracture.

B. Factors Affecting Sea Urchins' Population

Human-induced activities and ecological changes pose threats to populations of urchins in the sea. According to Lawrence, J.M. (2013) factors affecting sea urchin population are: overfishing disease, and changes in environment. Shears, and Evans, (2018), therefore, emphasize the negative effects of human-induced activities on sea urchin populations, in particular those areas experiencing immense pressure from fishing and pollution. The activities of individuals involved in sea urchin aquaculture can manipulate the population of sea urchins. Most of the culturists of sea urchins had considered culturing sea urchins as their major source of income rather than fishing,

according to Ungson, J. (2004). The sea urchin culture is situated in the shallow 9 seagrass beds and coral reef area in the village. In the Philippines, the most commercially exploited species of sea urchin is *Tripneustes gratilla*. The tropical species has been commercially gathered in the increasing number (Ungson, J. 2006). Anderson, S.C. et al. 2020 A modeling study "Reducing factory effluent releases significantly alleviate adverse impact on sea urchin populations" Factory effluents identify adverse effects on sea urchin populations that occur due to their impacts on spawning success, disease susceptibility, and mortality. Kim J.H. et al. 2018. It has been concluded in research conducted by Ohgaki et al. (2018) that altering the water temperature and the frequentation of algal blooms had significant impacts on sea urchin populations, resulting in abnormal larvae development. It would seem that sea urchin populations are somewhat more sensitive to human activity than previously thought, as evidenced in this vast half-century observational study.

C. Factors Affecting Sea Urchin Reproduction

Successful fertilization of sea urchin depends on factors such as population density and coordinated adult behavior during spawning. Instead, human activities negatively impact sea urchins, thus pushing their populations towards hazard while eradicating all their diversified habitat. These include overfishing, pollution, and coastal development, among others. These anthropogenic effects can also lead to changes in water quality and chemistry and thus affect the physiological as well as ecological functions of not only sea urchins but also their larvae (Leuchtenberger, S.G. et al 2018).¹⁰ Acidification is the reduction in pH of seawater through the uptake of excessive carbon dioxide in the atmosphere. Acidification threatens the calcification,

growth, and survival of both the sea urchins and their larvae. Furthermore, it can disrupt the fertilization process, impacting sperm motility, egg activation, and the development of sea urchin embryos (Dworjanyn and Byrne, 2018; Zhadan et al., 2017). Temperature variations significantly influence the gonadal development, gamete quality, fertilization success, and larval survival of sea urchins. The optimal temperature ranges for reproduction vary among different sea urchin species, shaped by their geographic distribution and adaptation (Zhadan et al., 2017). The desirable mass spawning of a certain number in a population is required to attain the concentrations of sperm necessary for fertilization, since synchronization in the timing of spawning is essential (Bronstein, O., et al., 2016). As such, besides population densities, the reproductive behavior is also involved in population effectiveness of fertilization, including adult aggregations at the time of spawning (Pennington, T. J. 1985). According to Brawley et al. (2019), sea urchins play key species in maintaining diversity of algae. The grazing pressure depresses the dominant alga species, which enables the understory algae to thrive. This grazing also aids nutrient regeneration to allow increased algal biomass. Moreover, sea urchins facilitate habitat complexity for several species of algae.

Research Gap

This research intensify to understanding of sea urchin population dynamics requires longitudinal population studies, especially with environmental changes and human impacts, where such studies have not yet been conducted in Gingoog City. Hall (2023) overviews sea urchins in terms of ecological roles and long-term monitoring requirements to determine the impact of environmental fluctuation on the species. The need for this is further emphasized by the Food and Agriculture Organization (FAO)

(2023) in its Global Sea Urchin Catches Report, which points out global trends in Sea urchin populations and underscores the importance of longitudinal studies to track changes resulting from fishing pressures and habitat degradation.

The Society of Integrative and Comparative Biology (SICB) (2017) also reports the different species of sea urchins in the Philippines and calls for continued monitoring as an integral part of conservation, especially in the face of human impacts. Cruz and Ebanez (2018) also report regional differences in sea urchin populations, which further supports the need for longitudinal studies to better understand these dynamics. Finally, Lyimo et al. (2011) discuss the habitat preferences of Collector Urchins (*Tripneustes gratilla*) and emphasize the need for long-term data collection to understand how environmental factors influence sea urchin populations over time. Collectively, these sources illustrate the pressing need for comprehensive longitudinal studies to monitor sea urchin populations in Gingoog City effectively.

The literature existing to date emphasizes the need for a more detailed environmental impact analysis regarding the distribution and morphometric traits of sea urchins. Lyimo et al. (2011) emphasize how habitat preferences of Collector Urchins (*Tripneustes gratilla*) are affected by environmental conditions such as water quality and substrate type, thus recommending targeted research into these factors. Tuya et al. (2007) further shows that the population dynamics of Black Sea Urchins (*Diadema antillarum*) is greatly influenced by the movement of water, which again underlines the role of particular environmental parameters. Cruz and Ebanez (2018) have also further emphasized the ecological role of sea urchin species in general found in the Philippines, noting gaps in knowledge regarding the impact of pollution and habitat degradation. The Reef Resilience Network (2023) and the FAO (2023) also note that alterations in

environmental conditions, including climate change and pollution, are known to impact sea urchin populations negatively and should be addressed with local studies. Altogether, these studies suggest that there is a dire need to study further the impact of specific environmental factors on the health and distribution of sea urchins.

MATERIALS AND METHODS

Research Design

This study aimed at investigating the morphometric characteristics of Sea urchins in two particular locations, specifically in Brgy. Agay-ayan and Brgy. San Luis. The researchers used a quadrat sampling method as the means of estimating the total number of sea urchins in each location. The collected data, which involved morphological measurements of the sea urchins, underwent statistical analysis with the help of SPSS Two-Way ANOVA, which included T-tests to evaluate the associations among these traits and the populations of sea urchins in the respective sites. This integrated process was conducted to understand more in detail the populations of sea urchins and the specific morphological variations that exist in the study areas.

Research Locale

The study was carried out in two different places in Gingoog City, namely, in Brgy. Agay-ayan and in Brgy. San Luis. Brgy. Agay-ayan has less corals, and less density of houses nearby (less than 10) with an elevation of 16.7 meters above sea level at coordinates (8.86° N, 125.1° E). On the other hand, Brgy. San Luis, with large boulders prominent, and close to a company, was seen at coordinates (8.88° N, 125° E), and 5.1 m above sea level. These varying environments thus offered a chance to examine the possible influence of habitat features on sea urchin populations.

Sampling Design

To estimate the total number of sea urchins within the study areas, researchers applied a quadrat sampling technique. For each site, four squares, each measuring 10 meters by 15 meters, were placed strategically 20 meters apart and parallel to the shoreline. All the sea urchins within each quadrat were then counted with detailed care and the count recorded to give estimates of the total population of sea urchins in different areas. Such methods allowed for an almost standardized and measurable estimation of sea urchin abundance within defined sampling areas.

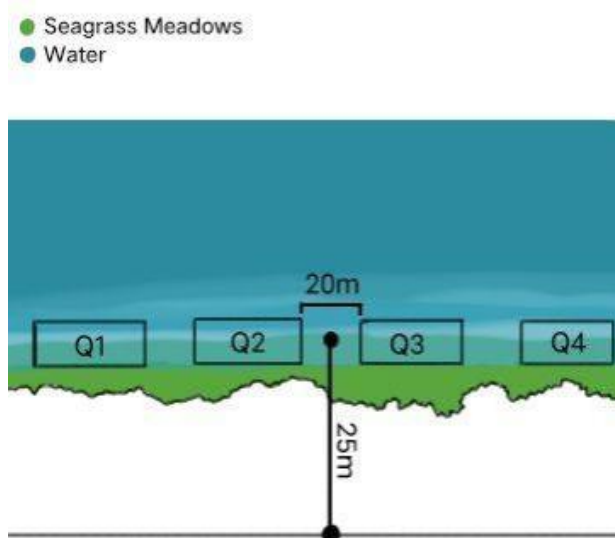


Figure 1: Quadrat – Placing in Brgy. Agay-ayan (without factory)

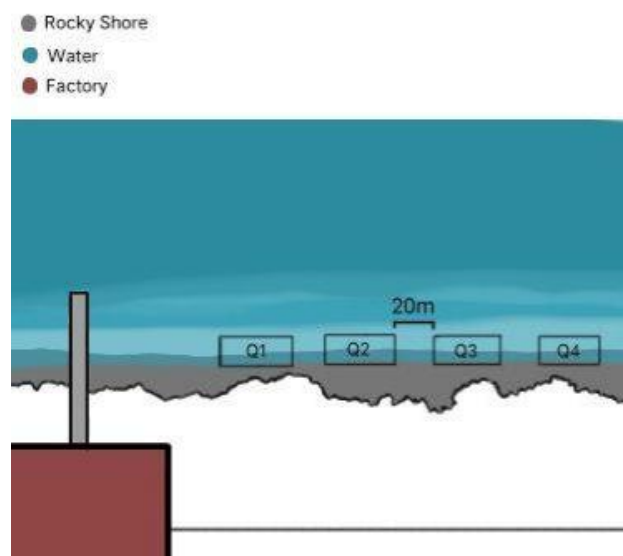


Figure 2: Quadrat – Placing in Brgy. San Luis (with factory)

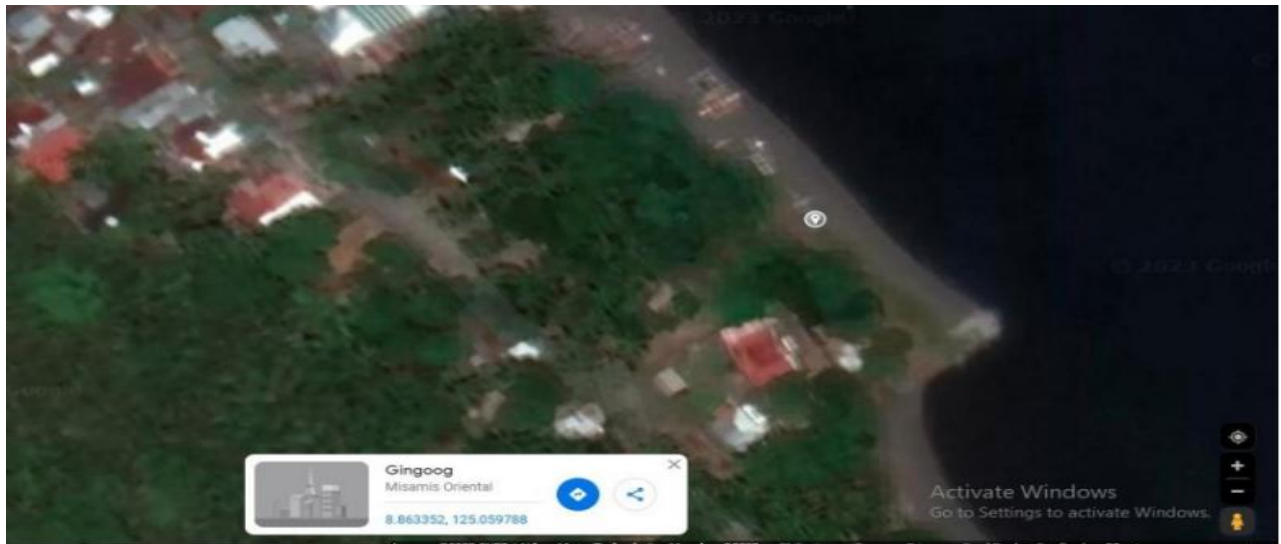


Figure 3: Brgy. Agay-ayan (without factory) Coordinates (Screenshot from Google Maps)

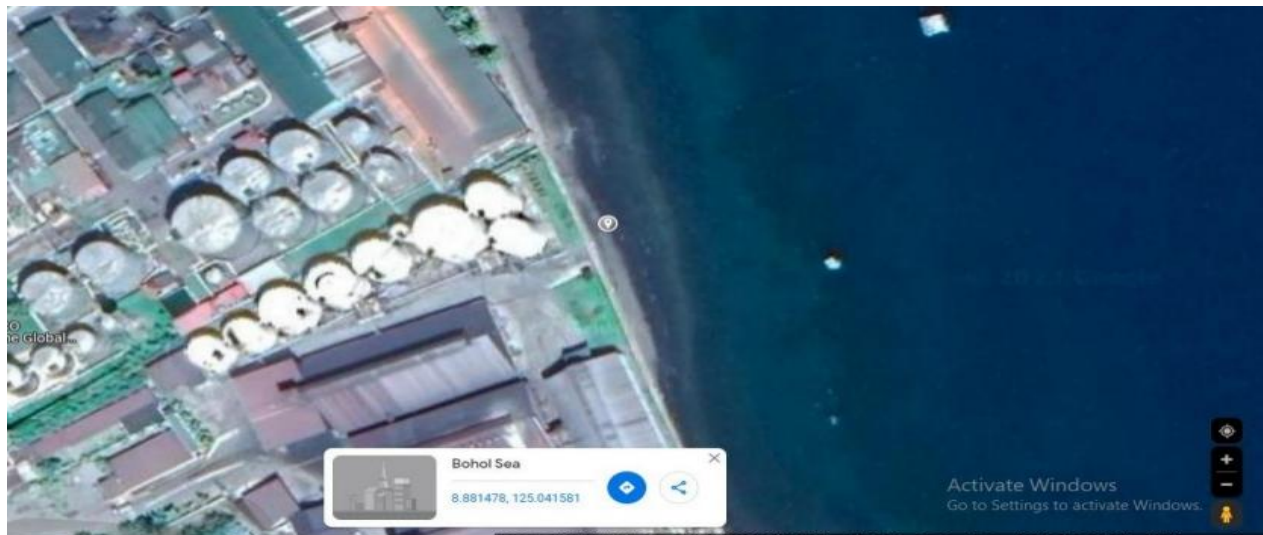


Figure 4: Brgy. San Luis (with factory) Coordinates (Screenshot from Google Maps)

Research Procedure

Sea urchin collections were carried out in a standardized and cautious manner. For our data collection during the month of October, we resolved to collect in the evening for every location since it coincided with low tide, and we could hence sample a more extensive range of species for population analysis. There were two buckets designated for the two species, any types of flashlight gadgets that are applicable for fishing in the evening, and stainless tongs for easily to hold the species: Collector Urchins (*Tripneustes gratilla*) Black Sea Urchin (*Diadema antillarum*). Sea urchins were hand-collected with a lot of care not to cause them any injuries. Following the collection, they were immediately transferred to the identified area for further inspection. Here, the sea urchin mass was weighed accurately, this time, using a calibrated digital scale which was reset to zero every time weighing ensued. For each sea urchin, the length of the longest spine was taken using a transparent ruler.

Research Analysis

We measured the weight, spine length, and the number of sea urchins in each location carefully. The data was then analyzed using techniques called Two-way ANOVA and T-tests. These tests helped us determine how the size and shape of the sea urchins were related to how many of them lived in each spot.

RESULTS AND DISCUSSION

Morphological Characteristics

Spine Length

Table 1. Descriptive Statistics on Morphological features of Collector Urchins (*Tripneustes gratilla*) and Black Sea Urchins (*Diadema antillarum*) in both Brgys. in terms of Spine Length.

Barangays	Species (Mean $\pm$ SD)	
	Black Sea Urchins (<i>Diadema antillarum</i>)	Collector Urchins (<i>Tripneustes gratilla</i>)
Agay-ayan	5.9624 $\pm$ 3.08934	.3916 $\pm$.32718
San Luis	6.1540 $\pm$.89491	.9550 $\pm$.63293

Table 1 displays the spine lengths of Black Sea Urchin (*Diadema antillarum*) in San Luis (mean = 6.1540) are slightly longer on average compared to those in Agay-ayan (mean = 5.9624). The variability in spine length is much greater in Agay-ayan (SD = 3.08934) than in San Luis (SD = 0.89491). This suggests that the spine lengths in San Luis are more consistent, while those in Agay-ayan show a wider range of values. While, the spine lengths in San Luis are more than twice as long on average (0.9550 cm) compared to those in Agay-ayan (0.3916 cm). Variability is slightly higher in San Luis (SD = 0.63293) than in Agay-ayan (SD = 0.32718), indicating greater differences in spine lengths among individuals in San Luis.

Researchers of this study have observed a significant difference in the spine length of sea urchins specifically Black Sea Urchin (*Diadema antillarum*) between two brgys. Sea urchins in Brgy. San Luis have longer spines compared to those in Brgy.

Agay-ayan. After thorough investigation, the researchers concluded that these differences in spine length are due to environmental factors that affect the health of the sea urchins. In Brgy. Agay-ayan, the seagrass ecosystem seems to negatively impact the health of Black Sea Urchin (*Diadema antillarum*), while Collector Urchins (*Tripneustes gratilla*) thrive in this environment. This was supported by Alwin Hylkema's (2015) that suggests Black Sea Urchin (*Diadema antillarum*) long-spined Sea urchin prefers coral reefs over sea grass habitats due to Ecological preferences food availability coral reefs provide abundant algae for grazing. Habitat complexity coral reefs offer shelter and protection. Water flow coral reefs experience optimal water flow for respiration.

In the study of Kasim,et.al (2009) he stated that sea urchin feeds on algae, small animals and sea grasses which the most prepared is Sea grass (*T. hemprichii*) over other sea grasses. In his further discussion he noted that most population of sea urchin in sea grasses is (*T. gratilla*) or collector sea urchin.

This was supported by Li Ji et al. (2021) factories often discharge both organic and inorganic nutrients into nearby water bodies. This can enhance nutrient availability, which is crucial for marine organisms like sea urchins. Organic nutrients can promote the growth of phytoplankton, a primary food source for many marine species, including sea urchins.

During in our data collection the researchers notice that the area of brgy. Agay-ayan have less of corals compared to brgy. San Luis, this was supported by (Waller, 1996) Black Sea Urchins (*Diadema antillarum*) has a highly developed light sensitivity.

They are mostly active during the night, during the day - they may hide in crevices within reef or corals.

Mass of Sea Urchins

Table 2. Descriptive Statistics on Morphological features of Collector Urchins (*Tripneustes gratilla*) and Black Sea Urchins (*Diadema antillarum*) in both Barangays in terms of Mass (g)

Barangays	Species (Mean $\pm$ SD)	
	Black Sea Urchins (<i>Diadema antillarum</i>)	Collector Urchins (<i>Tripneustes gratilla</i>)
Agay-ayan	.0781 $\pm$.09738	.0546 $\pm$.06200
San Luis	.0815 $\pm$.04277	.0795 $\pm$.03790

As shown in Table 2, Black Sea Urchins (*Diadema antillarum*) has a mean mass slightly higher in San Luis (0.0815 g) compared to Agay-ayan (0.0781 g), but the standard deviation is much lower in San Luis (0.04277) than in Agay-ayan (0.09738), indicating more consistent individual masses in San Luis. Compared to Collector Urchins (*Tripneustes gratilla*), the mean mass is higher in San Luis (0.0795 g) compared to Agay-ayan (0.0546 g). The standard deviation is also lower in San Luis (0.03790) compared to Agay-ayan (0.06200). This implies individuals in San Luis are not only heavier on average but also more uniform in size compared to Collector Urchins (*Tripneustes gratilla*) in Agay-ayan.

Across both brgys., Black Sea Urchins (*Diadema antillarum*) generally has higher mean mass compared to Collector Urchins (*Tripneustes gratilla*). This suggests that Black Sea Urchins (*Diadema antillarum*) individuals are larger on average than Collector

Urchins (*Tripneustes gratilla*). Longer spines and larger body size reflects increased access to food due to higher algal growth caused by nutrient runoff influenced by disposed organic wastes and nutrients like nitrogen and phosphorus. However, signs of stress, such as abnormal growth patterns or thinner spines, may appear due to pollution exposure.

Population of Analysis

Table 3. Descriptive Statistics on population of Collector Urchins (*Tripneustes gratilla*) and Black Sea Urchins (*Diadema antillarum*) in both brgys.

Barangays	Species (Mean $\pm$ SD)	
	Black Sea Urchins (<i>Diadema antillarum</i>)	Collector Urchins (<i>Tripneustes gratilla</i>)
Agay-ayan	9.38 $\pm$ 7.210	18.13 $\pm$ 20.511
San Luis	20.63 $\pm$ 5.449	3.13 $\pm$ 3.182

As observed in Table 3, the average population (9.38) of Black Sea Urchins (*Diadema antillarum*) in Agay-ayan is relatively low compared in San Luis (mean = 20.63), with a high standard deviation ($\pm$ 7.210), indicating significant variability in population counts within this barangay, while Black Sea Urchins (*Diadema antillarum*) in San Luis has lower standard deviation ($\pm$ 5.449), indicating more consistent population counts. The population of Black Sea Urchins (*Diadema antillarum*) is more abundant in San Luis than Agay-ayan, suggesting better environmental conditions, availability of resources, or reduced predation in San Luis.

Furthermore, Collector Urchins (*Tripneustes gratilla*) in Agay-ayan has higher average population (18.13), but the very high standard deviation (20.511) indicates extreme variability, potentially due to sampling inconsistencies or localized factors affecting population distribution. The average population of Collector Urchins (*Tripneustes gratilla*) is much lower (3.13) in San Luis, with a smaller standard deviation (± 3.182), showing more uniform but low counts. Collector Urchins (*Tripneustes gratilla*) is more abundant in Agay-ayan than in San Luis.

The higher population of Black Sea Urchins (*Diadema antillarum*) in San Luis indicates resilience to environmental stress, as this species is known for its ability to tolerate disturbed environments, such as those affected by industrial activities. The presence of factories creates nutrient runoff, increasing algal growth. The results of this study was supported by Andilala et al. (2020), sea urchins belonging to the (*Phylum Echinodermata*) are adapted to hard, benthic surfaces and are mainly algae consumers. Such organisms live well in high algal concentrations areas as these algae serve as their primary food source. Steinberg (2011) goes on to elaborate more on the interaction between sea urchins and algae in detail, especially emphasizing feedback mechanisms like herbivory and nutrient cycling that both contribute to creating such a fragile balance, which serves as food for Black Sea Urchins (*Diadema antillarum*).

Similar to what Matranga (2000) said, sea urchins possess a robust immune system that could help them survive from strong conditions this may include Black Sea Urchins (*Diadema antillarum*). Population dynamics are controlled by natural disturbances and fishing. During the week of data collection in Brgy. Agay-ayan, local

residents were seen collecting mature Black Sea Urchins (*Diadema antillarum*). Adult sea urchins play an important role in reef ecology because of their grazing nature (Williams, 2022). These may result in high numbers of this species, thus significant ecological impacts like dominance of large algae, which can smother corals (McClanahan & Shafir, 2000). In addition, anthropogenic activities, such as industrial effluents, can promote algal growth and lead to larger sea urchins in affected areas, although pollution may degrade habitats, negatively impacting populations (Prado and Galvez, 2008). Overall, these factors point out the resilience of sea urchins while also emphasizing the fragility of their ecosystems. The lower population of Black Sea Urchins (*Diadema antillarum*) could be due to less algal growth in undisturbed waters, as algae production is typically higher in areas with increased nutrient levels.

Figures 5 and 6 gives additional evidence that Black Sea Urchins (*Diadema antillarum*) in San Luis has greater population than in Agay-ayan, while Collector Urchins (*Tripneustes gratilla*) in Agay-ayan has greater population than San Luis.

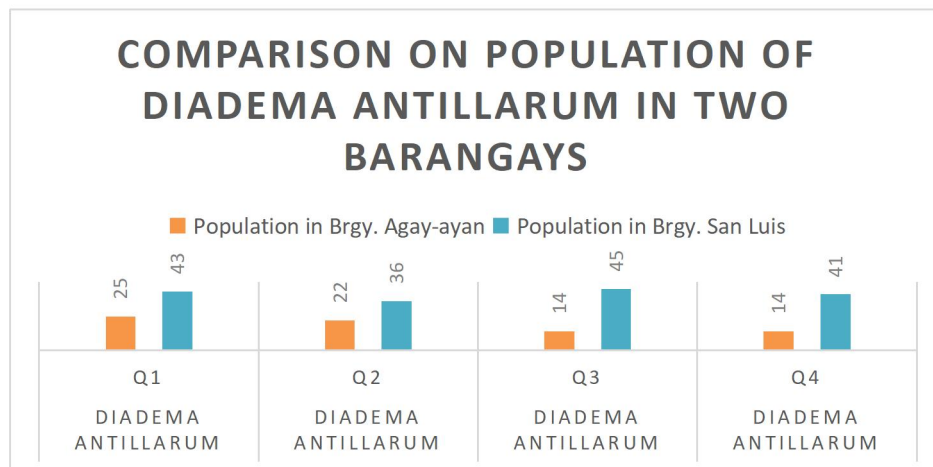


Figure 5. Graphical representation of comparison on population Of Black Sea Urchins (*Diadema antillarum*) in both Brgys.

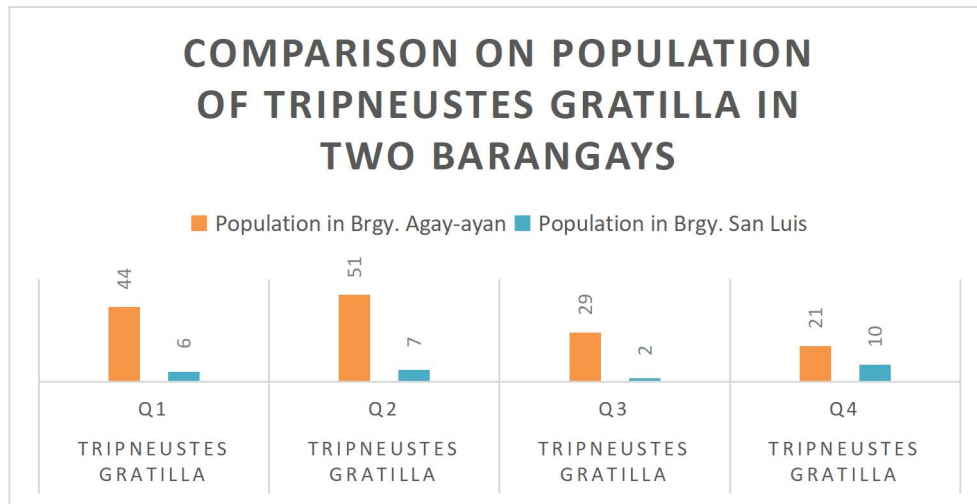


Figure 6. Graphical representation of comparison on population Of Collector Urchins (*Tripneustes gratilla*) in both Brgys.

The findings were corroborated by Santos et al. (2018), who investigated the influence of human activities and adjoining factory on the population of Collector Urchins (*T. gratilla*) in the Ria de Aveiro estuary, Portugal. Their result revealed a significance decrease in Collector Urchins (*T. gratilla*) population in areas with heightened human activity and the disturbance closer proximity to the factory. They proposed that the these effects stem from substantial human impacts, potentially inducing physiological stress in the Collector Urchins (*T. gratilla*). On the other hand, Brgy. San Luis, located near a coconut oil factory, experiences nutrient runoff into the water, this nutrient influx has boosted the marine ecosystem in that area. Additionally, the lack of corals and rocks, coupled with human activities, in Brgy. Agay-ayan could be contributing factors to the observed differences in the spine length of Black Sea Urchins (*D. antillarum*).

Table 4. Independent Samples Test results on Population and Morphological features of the species in both Brgys.

Source	Barangays	df	Mean Difference	t	Sig.
Population	Agay-ayan	30	1.875	.406	.688
	San Luis	25.638	1.875	.406	.688
Spine Length	Agay-ayan	30	-.37750	-.333	.741
	San Luis	28.316	-.37750	-.333	.742
Mass	Agay-ayan	30	-.01312	-.594	.557
	San Luis	21.441	-.01312	-.594	.559

Note: *Significant at 0.05 level of significance; $p > 0.05$ implies not significant.*

Based on table 4, on population of the two species the p-value (0.688) is greater than 0.05 significance level, indicating no statistically significant difference in population between Agay-ayan and San Luis. This suggests that the total population of the species is similar across the two brgys. The p-value (0.741) on spine length of species across locations is also greater than 0.05, indicating no statistically significant difference in spine length between the two brgys. The negative mean difference suggests that species in San Luis have slightly longer spines on average, but this difference is not meaningful statistically. Furthermore, the p-value (0.557) on mass (g) of species across brgys. is also greater than 0.05, indicating no statistically significant difference in mass between the two brgys. The negative mean difference suggests that species in San Luis are slightly heavier, but the difference is negligible and not significant.

A graphical representation (bar graph) of the comparison on total population of Collector Urchins (*T. gratilla*) and Black Sea Urchins (*D. antillarum*) in both Brgys. and box-plot of the comparison on morphological features of Collector Urchins (*T. gratilla*)

and Black Sea Urchins (*Diadema antillarum*) in both brgys. are displayed below to support the evidence in table 4.

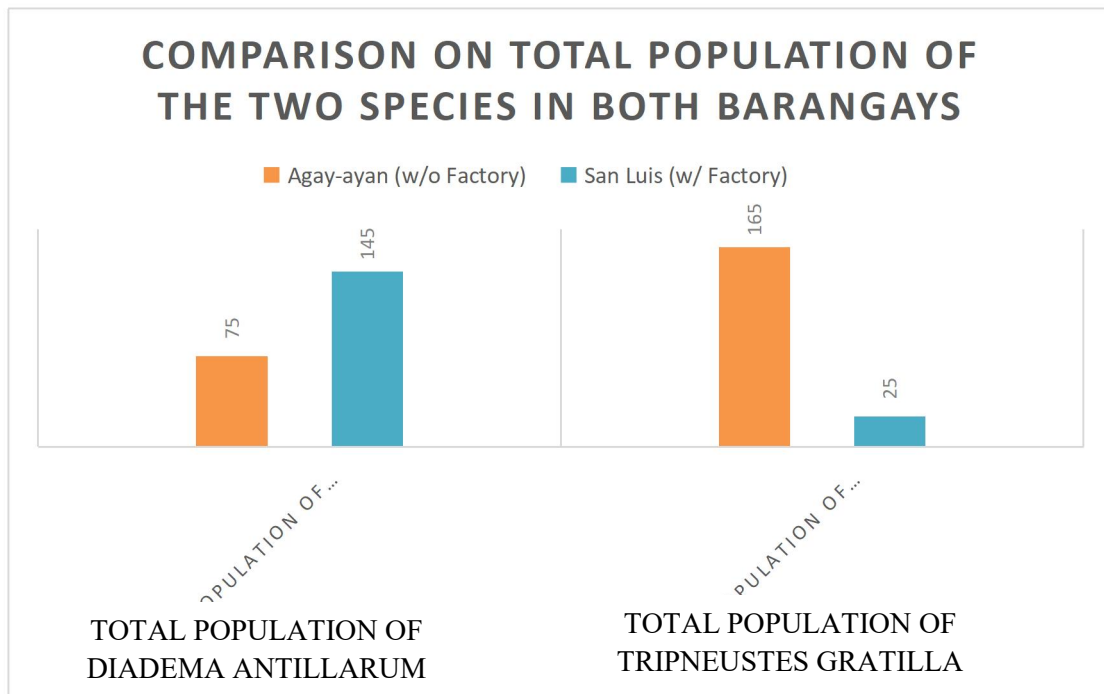


Figure 7. Graphical representation of comparison on total population of (*Tripneustes gratilla*) and (*Diadema antillarum*) in both Brgys.

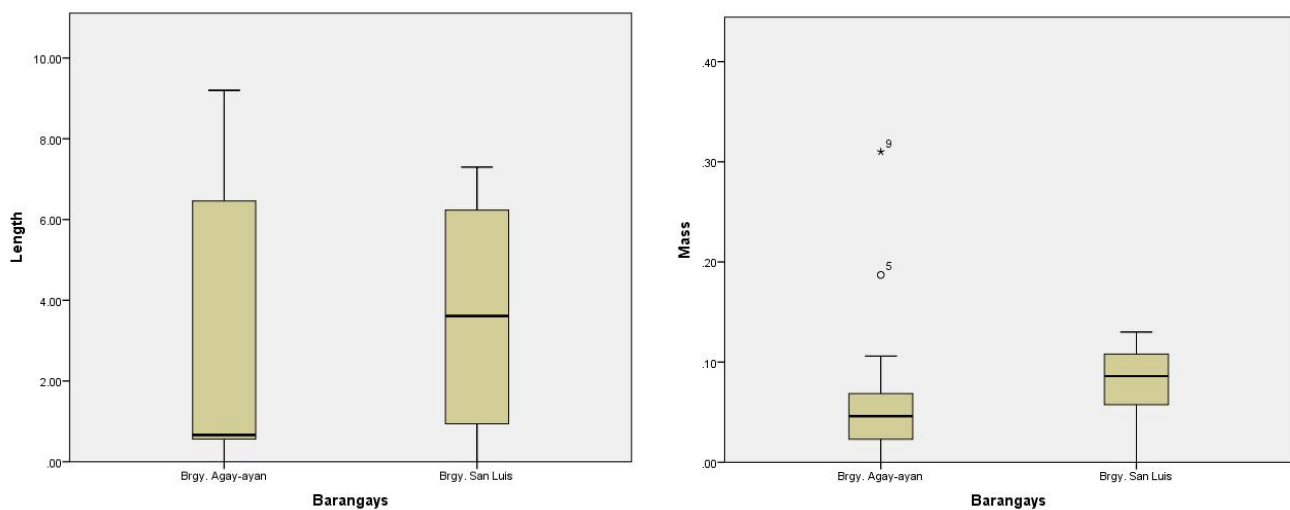


Figure 8. Box-plot chart representation of comparison on morphological features of (*Tripneustes gratilla*) and (*Diadema antillarum*) in both Brgys.

The results in Table 4 shows the Two-Way ANOVA on population and morphometric features of Collector Urchins (*T. gratilla*) and Black Sea Urchins (*D. antillarum*) when compared between brgys., species, and the interaction between brgys. and species.

Table 4a. Two-Way ANOVA Test results on Population and Morphological features of Collector Urchins (*T. gratilla*) and Black Sea Urchins (*D. antillarum*) in both Brgys.

Source	Dependent Variables	df	Mean Square	F	Sig.
Barangays	Population	1	28.125	.220	.643
	Spine Length (cm)	1	1.140	.420	.522
	Mass (g)	1	.001	.336	.567
Species	Population	1	153.125	1.195	.284
	Spine Length (cm)	1	231.975	85.501	.000
	Mass (g)	1	.001	.186	.670
Barangays * Species	Population	1	1378.125	10.756	.003
	Spine Length (cm)	1	.276	.102	.752
	Mass (g)	1	.002	.369	.548

Note: Significant at 0.05 level of significance; $p > 0.05$ implies not significant.

The effect of Barangays having a p-values for all three dependent variables (Population, Spine Length, Mass) are greater than 0.05, indicating no significant differences in population, length, or mass between the two brgys. This suggests that the location (with or without factory infrastructure) does not significantly influence these variables. For the effect of species, the non-significant results for population ($p = 0.284$) indicate that the total population is comparable between Collector Urchins (*T. gratilla*) and Black Sea Urchins (*D. antillarum*). The significant result for length ($p = 0.000$) suggests that there is a meaningful difference in spine length between the two species. This is expected since the two species have distinct morphologies, with Black Sea

Urchins (*D. antillarum*) having much longer spines than Collector Urchins (*T. gratilla*). The non-significant results for mass ($p = 0.670$) indicate that the body mass of the two species is not statistically different. While, the interaction effect of Brgys. and Species, shows a significant interaction for population ($p = 0.003$) indicating that the combined effect of barangay location and species is important. This means that the population distribution of the two species varies between Agay-ayan and San Luis in a way that depends on the species and the non-significant interactions for length ($p = 0.752$) and mass ($p = 0.548$) suggest that the differences in spine length and mass between species do not depend on the brgy.

The lack of significant differences across all variables (population, length, and mass) for Brgys. implies that the presence of factory infrastructure in San Luis does not exert a measurable effect on the population size or morphology of these species. This indicates that other ecological factors (limited to this study), such as water quality or habitat type, are relatively consistent between the two locations. The significant difference in spine length between the two species highlights a natural variation in morphology, which aligns with known species-specific traits. The non-significant differences in population and mass suggest similar ecological roles or adaptive capacities of Collector Urchins (*T. gratilla*) and Black Sea Urchins (*D. antillarum*) in the studied locations. The significant interaction for population suggests that the species respond differently to the two brgys.

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary of the Results

Morphometric and Population Analysis of Black Sea Urchin (*Diadema antillarum*) and Collector Urchins (*Tripneustes gratilla*) in Barangay Agay-ayan and San Luis, Gingoog City, Philippines, by Gingoog City United Colleges' students, sought to identify the effect of a factory in Barangay San Luis on sea urchin populations. Quadrat sampling was used to compare the spine length, mass, and population of the two species in the two sites: Barangay Agay-ayan (factory-free) and Barangay San Luis (factory-present). Although the factory had no direct significant impact on the total population, spine length, or mass of the species, it did have an effect on species-specific responses, especially in population distribution. Black Sea Urchins (*D. antillarum*) were present in greater numbers in San Luis, possibly due to more algal growth brought about by excess nutrient runoff, whereas Collector Sea Urchins (*T. gratilla*) were more numerous in Agay-ayan, indicating a preference for less disturbed environments. The research indicates the need to take into consideration species characteristics and local environmental conditions when assessing anthropogenic influences on marine ecosystems.

Conclusions

The study examined the population and morphological features (spine length and mass) of Collector Urchins (*Tripneustes gratilla*) and Black Sea Urchins (*Diadema antillarum*) in two brgy.s. namely: Agay-ayan and San Luis to determine the influence of environmental factors, including the presence of factory infrastructure.

The statistical results revealed no significant difference in total population between the two barangays ($p > 0.05$), although Black Sea Urchins (*D. antillarum*) was more abundant in San Luis and Collector Urchins (*T. gratilla*) in Agay-ayan. However, a significant interaction ($p = 0.003$) suggests that population distribution is species-specific and influenced by local environmental factors. A significant difference ($p = 0.000$) in spine length was found between the species, reflecting their natural morphological traits, with Black Sea Urchins (*D. antillarum*) having longer spines. However, spine length was not significantly influenced by brgy. location ($p = 0.741$). There were also no significant differences in mass between brgys. or species ($p > 0.05$), indicating similar body sizes across locations. San Luis, characterized by factory infrastructure, exhibited higher Black Sea Urchins (*D. antillarum*) populations, potentially due to increased algal growth from nutrient runoff. This species is known for its resilience in disturbed environments. Conversely, Collector Sea Urchins (*T. gratilla*) was more abundant in Agay-ayan, suggesting that this species prefers less disturbed habitats with better overall environmental quality.

The presence of factories in San Luis did not significantly affect the overall population, spine length, or mass of the species but likely altered resource availability (algal growth) that favored Black Sea Urchins (*D. antillarum*). However, long-term monitoring is needed to assess potential stress, such as abnormal growth or habitat degradation, due to pollution.

The results indicate that while the factory infrastructure in San Luis has not caused measurable differences in population or morphology, it may influence species-specific responses, particularly in terms of population distribution. The findings

underscore the importance of considering species traits and local environmental factors when evaluating anthropogenic impacts on marine ecosystems.

Recommendations

The recommendation aims to provide guidance and suggestions for future actions or decisions based on the study's findings and conclusions related to the comparison of morphometric and population analysis. This study recommends to include measurement of shell width along with length and mass in the morphometric analysis of sea urchins following standardized measurement techniques across sites for comparable data. This suggestion aims to enhance the understanding of sea urchins morphology. Additionally, to better understand the breeding ecology of Sea urchin populations in the study areas, it is advised to assess reproductive parameters. This includes examining gonad size, maturity, and spawning patterns, providing knowledge on the reproductive dynamics of these sea urchin populations. Considering the ecological focus of the study, it is also recommended to assess how human activities, such as pollution or coastal development, might affect sea urchin populations. Consistent measurements, along with attention to reproductive factors and human-induced stressors, will make the study's findings more reliable and thorough.

Future studies should then be longitudinal studies to track sea urchin populations in Gingoog City, specifically on the effect of certain environmental variables (water quality, substrate) on sea urchin condition and distribution. Additional studies must also involve a more comprehensive environmental impact assessment, studying the effects of pollution and habitat destruction on sea urchin populations, and investigating the reproductive characteristics of these species to gain a better understanding of their

population dynamics. Lastly, standardized measurement methods must be established and implemented in various marine habitats to provide comparable data for future research.

For farmers, the most useful variable from this research is that Collector Urchins (*Tripneustes gratilla*) do well in less disturbed waters, and Black Sea Urchins (*Diadema antillarum*) are resilient in waters with runoff of nutrients (such as near factories).

This implies that sea urchin harvesting methods in the future should take into account local environmental conditions around varying coastlines. Farmers where man impacts are less and water quality is improved ought to emphasize sustainably harvesting Collector Urchins (*Tripneustes gratilla*), whereas farmers where there is more nutrient richness may look to Black Sea Urchins (*Diadema antillarum*) as a possible, though perhaps less quality, harvest option. More studies should be conducted to ascertain the long-term impacts of nutrient runoff on the quality and safety of Black Sea Urchins (*Diadema antillarum*) for consumption. The research underscores the significance of comprehending the interaction between environmental conditions and sea urchin populations for sustainable and effective farming practice.

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CHRISTIAN GUILLEMORE O. SUSI
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Bachelor of Science in Agribusiness

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Chapter 3- Materials and Methods	%	%
Chapter 4- Results and Discussion	%	%
Chapter 5-Summary, Conclusion & Recommendations	%	%
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CHRISTIAN GUILLEMORE O. SUSI
Research Adviser APPENDIX

A. Raw Data

Table 1a. Descriptive Statistics on Morphological features of Collector Urchins (*T. gratilla*) and Black Sea Urchins (*D. antillarum*) in both Brgys. between quadrats

Barangays	Quadrat (Mean $\pm$ SD)				Total Mean	Total SD
	Quadrat 1	Quadrat 2	Quadrat 3	Quadrat 4		
Brgy. Agay-ayan	3.87 $\pm$ 4.024	4.18 $\pm$ 4.457	3.46 $\pm$ 4.214	1.20 $\pm$ 1.674	3.1770	3.57482
Brgy. San Luis	4.04 $\pm$ 2.835	3.33 $\pm$ 2.900	3.25 $\pm$ 3.466	3.60 $\pm$ 3.142	3.5545	2.78722

B. DOCUMENTATION



Figure 9. Brgy. San Luis Sampling (Site 1)



Figure10. Brgy. Agay-ayan Sampling (Site 2)



Figure 11. Collected specimen of Black Sea Urchin (*Diadema antillarum*)



Figure 12. Weighing the specimen of Collector Urchin (*Tripneustes gratilla*)



Figure 13. Collector Urchins (*Tripneustes gratilla*)



Figure 14. Black Sea Urchins (*Diadema antillarum*)

C. SAMPLE LOG BOOK ENTRY

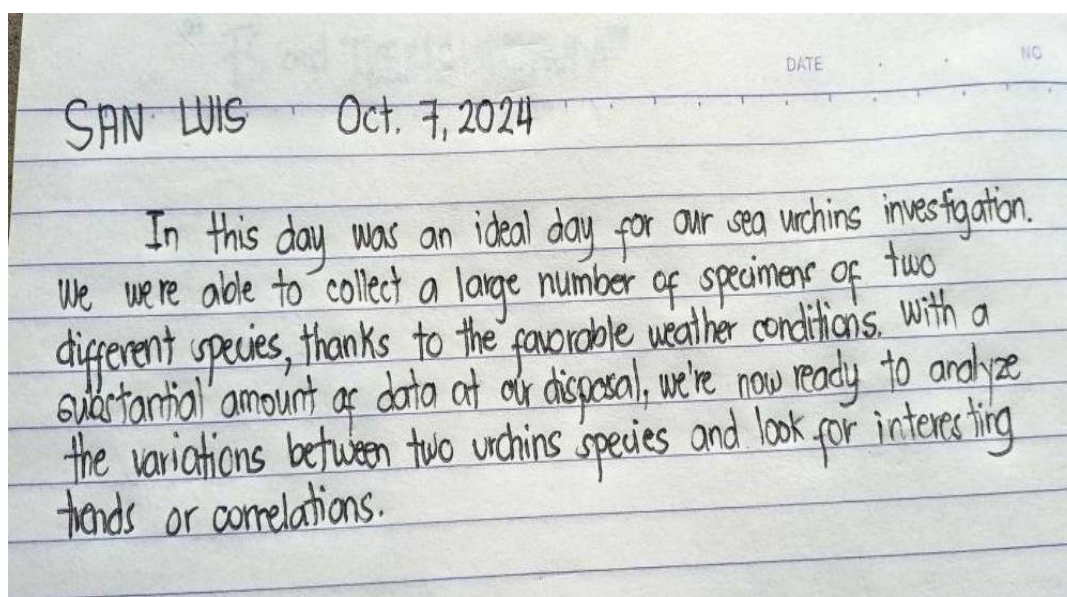


Figure 15. First day of Brgy. San Luis

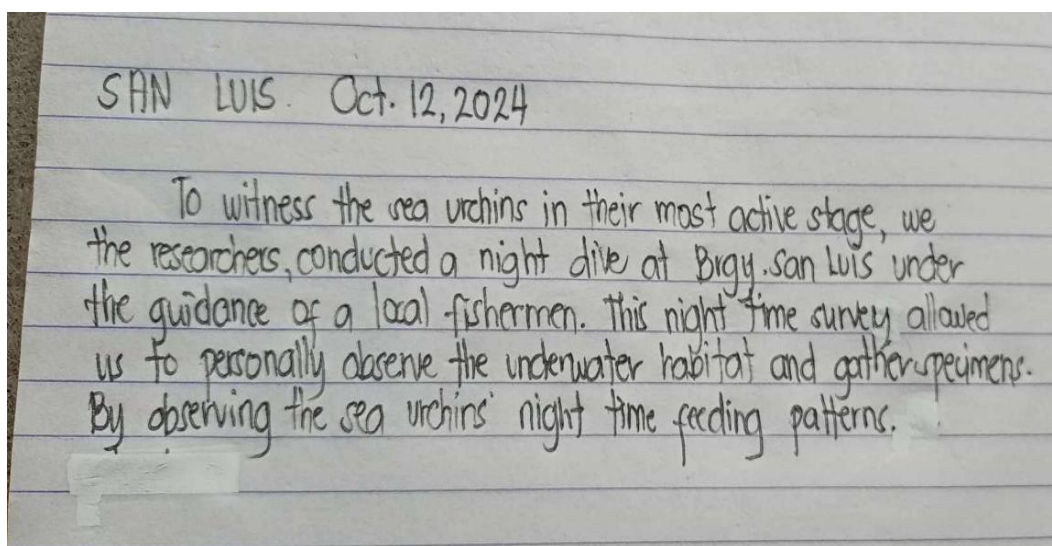


Figure 16. Second Day of Brgy. San Luis

AGAY-AYAN Oct. 8, 2024

During our visit to Brgy. Agay-ayan, we followed a similar approach to our work in Brgy. San Luis. We established four 10x15 quadrats and collected sea urchins from each. The quantity of sea urchins varied between quadrats. To further analysis, we measured the mass and spine length of each collected specimen.

Figure 17. First Day of Brgy. Agay-ayan

AGAY-AYAN Oct. 13, 2024

On our final day in Brgy. Agay-ayan, we spent the afternoon collecting and observing sea urchins. The dominant species we encountered were collector sea urchins (*Triplaster gratilla*), with fewer black sea urchins (*Diadema antillarum*) present. Our findings on this day aligned with our initial observations, reinforcing the patterns we had identified in our research.

Figure 18. Second Day of Brgy. Agay-ayan

BIOGRAPHICAL DATA

Personal Data



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Email Add: charlinecaitom@gmail.com
Cellphone no.: 0991-721-2921
Name of Father: Rogelio E. Caitom
Name of Mother: Helen R. Caitom

School Graduated and Honors:

Elementary: Roy Vizcara Elementary School
High School: Lurisa National High School
College: Bachelor of Science in Agribusiness, 2024-2025
Gingoog City United Colleges

BIOGRAPHICAL DATA

Personal Data



Name: **JUVELYN V. DAHANG**
Address: Purok 3 Boyugon, Duka, Medina, Misamis, Oriental
Birthday: January 19, 2002
Zodiac Sign: Capricorn
Place of Birth: Purok 3, Barangay Duka, Medina, Misamis, Oriental
Civil Status: Single
Religion: Roman Catholic
Email Add: juvelyndahang502@gmail.com
Cellphone no.: 0992-489-5129
Name of Father: Julierto Oclarit Dahang
Name of Mother: Rogelyn Citoy Vergara

School Graduated and Honors:

Elementary: Bulwa Elementary School
High School: Medina National Comprehensive High School
College: Bachelor of Science in Agribusiness, 2024-2025
Gingoog City United Colleges

BIOGRAPHICAL DATA

Personal Data



Name: **MAHARLIKA ANTONIA R. DEROCA**
Address: Ilang-ilang, AnakanGingoog City
Birthday: June 13, 2001
Zodiac Sign: Gemini
Place of Birth: Emergency Hospital Gingoog city
Civil Status: Single
Religion: Roman Catholic
Email Add: maharlikaantoniaroaderoca@gmail.com
Cellphone no.: 0991-610-6817
Name of Father: Elmer Derooca
Name of Mother: Marivic Roa

School Graduated and Honors:

Elementary: Anakan Elementary School
High School: Anakan National Comprehensive High School
College: Bachelor of Science in Agribusiness, 2024-2025
Gingoog City United Colleges

BIOGRAPHICAL DATA

Personal Data



Name: **LOWIE VICENTE GALUPO**
Address: Purok 3 Berenesan, Barangay Kianlagan Gigoog City
Birthday: June 20, 2001
Zodiac Sign: Libra
Place of Birth: Purok 3, Barangay 19 Gingoog City
Civil Status: Single
Religion: Roman Catholic
Email Add: galowizegalupo@gmail.com
Cellphone no.: 0953-538-4947
Name of Father: Faustino A. Galupo
Name of Mother: Elvira V. Galupo

School Graduated and Honors:

Elementary: Kianlagan Elementary School
High School: KISANDI National High School
College: Bachelor of Science in Agribusiness, 2024-2025
Gingoog City United Colleges

BIOGRAPHICAL DATA

Personal Data



Name: **JAYSAN RAZO**
Address: Purok 1,Barangay Lunotan
Birthday: July 29, 2001
Zodiac Sign: Leo
Place of Birth: Purok 1, Barangay LunotanGingoog City
Civil Status: Single
Religion: Roman Catholic
Email Add: jaysanpaygalan@gmail.com
Cellphone no.: 0992-489-5129
Name of Father: Nilo S. Paygalan
Name of Mother: Lordgena R. Paygalan

School Graduated and Honors:

Elementary: Lunotan Elementary School
High School: Lurisa National High School
College: Bachelor of Science in Agribusiness, 2024-2025
Gingoog City United Colleges

BIOGRAPHICAL DATA

Personal Data



Name: **JADE IVEN N. TEROL.**
Address: P-3 Brgy 24 Gingoog city
Birthday: May 06, 2002
Zodiac Sign: Taurus
Place of Birth: Purok 3, Barangay 24
Civil Status: Single
Religion: Roman Catholic
Email Add: jadeiventerol@gmail.com
Cellphone no.: 0965-975-7961
Name of Father: N/A
Name of Mother: Helen Neri Terol

School Graduated and Honors:

Elementary: Magallanes Elementary School
High School: Gingoog City Comprehensive National High School
College: Bachelor of Science in Agribusiness, 2024-2025
Gingoog City United Colleges

BIOGRAPHICAL DATA

Personal Data



Name: **JHON PET VARQUEZ**
Address: Purok 1 Baybay Lunao, Gingoog City
Birthday: June 29, 2003
Zodiac Sign: Cancer
Place of Birth: Purok 1 Baybay Lunao, Gingoog City
Civil Status: Single
Religion: Roman Catholic
Email Add: varquezjhonpet05@gmail.com
Cellphone no.: 0907-876-8395
Name of Father: N/A
Name of Mother: Jessica Varquez Borja

School Graduated and Honors:

Elementary: Baybay Lunao Elementary School
High School: Lunao National High School
College: Bachelor of Science in Agribusiness, 2024-2025
Gingoog City United Colleges

